UAV Firmware Anomaly Detection

Exploratory firmware screening report

Analysis result

Field	Value
Filename	arducopter (10).apj
Family	ArduPilot
Verdict	NORMAL
Anomaly score	0.307407662
Decision threshold	0.333895943
Analysis time (UTC)	2026-10-14T11:40:42.143206+00:00
Model ID	ocsvm-66e6524bb254
Board ID	58
Build identity	92b0cd78
Main flash bytes	1546180
External flash bytes	0
Candidate / retained features	464 / 433

The anomaly score does not exceed the model threshold. This determines the reported verdict. Scores are not probabilities of compromise.

Evidence identification

Original-container SHA-256:
64a8e2c2583aa3efb5736107655c25a873418d3c36f1e85fff7eb15788a2430a

Complete-payload SHA-256:
7f4a01b79c9e25ed0ffff79f441fccb331bdf08edc267e5fb157ed49226e3cb1

Model-file SHA-256:
66e6524bb254b6d945fbb329c45078ccdf3f28a70226e95f4921d26def3f388d

Supporting feature observations

Individual feature deviations from training medians. These are supporting observations, not causal model attribution.

Feature	Observed	Training median	Departure / SD
byte_frequency_203	0.00121202	0.000834407	4.566
window_entropy_max	7.75868	7.95111	-4.391
byte_frequency_036	0.00444968	0.00227834	3.297
byte_frequency_206	0.00103222	0.000612372	3.288
byte_frequency_024	0.0058292	0.0040191	2.938

Feature	Observed	Training median	Departure / SD
byte_frequency_124	0.00102381	0.00124776	-2.757
hashed_bigram_018	0.0088308	0.00678984	2.719
hashed_bigram_126	0.00620433	0.00584444	2.542

UAV static content evidence

Matches identify selected flight-stack content. They do not establish enabled settings, correct implementation or runtime behaviour. Offsets are relative to the named decoded flash region.

Flight Control

Marker	Count	Example evidence
attitude_parameters	10	ATC_RAT_YAW_FILT [main offset 1343544]; ATC_INPUT_TC [main offset 1343763]; ATC_RAT_RLL_FLTD [main offset 1343776]
position_parameters	12	PSC_VELXY_P [main offset 1343561]; PSC_VELXY_I [main offset 1343573]; PSC_VELXY_IMAX [main offset 1343585]
motor_parameters	1	MOT_TLT [main offset 1440173]

Navigation RTL

Marker	Count	Example evidence
rtl_parameters	10	RTL_ALT_TYPE [main offset 1324242]; RTL_ALT [main offset 1324310]; RTL_OPTIONS [main offset 1344717]
smart_rtl_parameters	2	SRTL_RTL [main offset 1439325]; SRTL_LAND [main offset 1439334]
mission_parameters	2	MIS_RST [main offset 1439602]; MIS_ITEMS [main offset 1457455]

Arming Failsafe

Marker	Count	Example evidence
arming_parameters	1	ARMING_CHECK [main offset 1344207]
prearm_messages	1	PreArm [main offset 1457528]
flight_failsafe_parameters	8	FS_THR_VALUE [main offset 1324094]; FS_GCS_ENABLE [main offset 1324107]; FS_EKF_FILT [main offset 1344437]

Geofence

Marker	Count	Example evidence
fence_parameters	12	FENCE_FETCH_POINT [main offset 1416960]; FENCE_RADIUS [main offset 1434067]; FENCE_MARGIN [main offset 1434093]
geofence_references	1	Geofence [main offset 1449106]
fence_breach_messages	1	Fence breach [main offset 1350545]
fence_recovery_messages	1	fence recovery [main offset 1350707]

State Estimation

Marker	Count	Example evidence
ek3_parameters	15	EK3_MAG_CAL [main offset 1389124]; EK3_SRC1_YAW [main offset 1389140]; EK3_GPS_TYPE [main offset 1389236]
ekf_diagnostics	5	EKF compass [main offset 1324479]; EKF position [main offset 1324500]; EKF velocity [main offset 1324522]

Interpretation and limitations

Exploratory model; detection performance remains limited.

A normal result does not establish authenticity or absence of malware.

An anomaly indicates statistical deviation, not proof of maliciousness.

UAV marker presence does not establish enabled settings or runtime behaviour.

New boards, releases and build configurations may differ legitimately.

Training provenance is not verified for every candidate sample.